the degeneracy pressure is so tiny that one never notices it, but at the huge densities of white dwarfs it is enormous.

Box 4.1

A Brief History of Wave/Particle Duality

Already in Isaac Newton's time (the late 1600s), physicists were struggling over the issue of whether *light* is made of particles or waves. Newton, though equivocal about the issue, leaned toward particles and called them *corpuscles,* while Christiaan Huygens argued for waves. Newton's particle view prevailed until the early 1800s, when the discovery that light can interfere with itself (Chapter 10) converted physicists to Huygens' wave viewpoint. In the mid-1800s, James Clerk Maxwell put the wave description on a firm footing with his unified laws of electricity and magnetism, and physicists then thought the issue had finally been settled. However, that was before quantum mechanics.

In the 1890s Max Planck noticed hints, in the shape of the spectrum of the light emitted by very hot objects, that something might be missing in physicists' understanding of light. Einstein, in 1905, showed what was missing: Light sometimes behaves like a wave and sometimes like a particle (now called a *photon).* It behaves like a wave, Einstein explained, when it interferes with itself; but it behaves like a particle in the *photoelectric effect:* When a dim beam of light shines on a piece of metal, the beam ejects electrons from the metal one at a time, precisely as though individual particles of light (individual photons) were hitting the electrons and knocking them out of the metal's surface one by one. From the electrons' energies, Einstein inferred that the photon energy is always inversely proportional to the light's wavelength. Thus, the photon and wave properties of light are intertwined; the wavelength is inexorably tied to the photon energy. _Einstein's discovery of the wave/particle duality of light, and the tentative quantum mechanical laws of physics that he began to build around this discovery, won him the 1921 Nobel Prize in 1922.

Although Einstein almost single-handedly formulated general relativity, he was only one among many who contributed to the laws of quantum mechanics—the laws of the "realm of the small."